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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Digital Document Record Management System

Submitted by

SWETHA K

25BCS158

Under the Guidance of

Mr ARAVIND

Technical Trainer

Technical Department, KG College of Arts and Science

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DECLARATION

I hereby declare that the project entitled “Digital Document Record Management System” submitted in partial fulfillment of the requirements for the course Java programming is a **record of my original work** carried out under the guidance of **Mr Aravind** Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

_____SWETHA K_____

CERTIFICATE

This is to certify that the project entitled **"Digital Document Record Management System"** is a **bona fide work** carried out by Swetha k **I -'B'**, BSC Computer Science, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java programming**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:10/4/26

Submitted for the Viva-Voce Examination held on _____10/4/2026_____

Guide:
(Technical Trainer, KGCAS)

_____Mr Aravind_____

Head of the Department
(Technical Head, KGCAS)

___Mr BoopalanSundaramoorthy___

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ABSTRACT

The **Digital Document Record Management System (DDRMS)** is a Java-based application developed to store, manage, and organize documents in a digital format. The main objective of this system is to replace traditional paper-based record keeping with an efficient electronic document management solution.

In many organizations, maintaining physical documents leads to problems such as loss of files, storage difficulties, time-consuming searches, and security risks. This project provides a computerized solution that allows users to upload, store, search, update, and delete documents easily through a user-friendly interface.

The system is developed using Java and uses file handling and database concepts to securely store document details. It provides features like user authentication, document categorization, quick search functionality, and secure access control to ensure data safety and privacy.

This project helps improve productivity, reduces paperwork, saves storage space, and ensures fast retrieval of important records. The Digital Document Record Management System can be used in schools, colleges, offices, hospitals, and other organizations for efficient document handling.

1.Introduction

The Digital Document Record Management System is a software application designed to store, manage, and retrieve documents in a digital format. In many organizations, handling physical documents can lead to issues such as data loss, misplacement, and difficulty in accessing records. This system provides an efficient and secure way to organize documents electronically.

The system allows users to add, update, search, and manage records with ease. It ensures that important documents are stored systematically and can be accessed quickly whenever needed. By digitizing records, the system reduces paperwork, saves time, and improves overall productivity.

This project is developed using programming concepts such as Object-Oriented Programming (OOP) and data structures to manage document information effectively. It also introduces users to real-world applications of software in areas like administration, business, and education.

Overall, the Digital Document Record Management System helps in maintaining data accuracy, improving security, and simplifying document handling processes.

1.1 Objective

To develop a paperless document system
Replace manual file storage with a digital system that stores documents electronically.

- **To provide easy document storage and retrieval**
  Allow users to upload, search, view, and access documents quickly using keywords.
- **To ensure document security and privacy**
  Implement user login and access control to protect confidential records.
- **To allow document management operations**
  Enable users to add, update, delete, and organize documents efficiently.
- **To improve productivity and reduce storage space**
  Save time, reduce paperwork, and minimize physical storage requirements.

1.2 Overview

The **Digital Document Record Management System (DDRMS)** is a Java-based software application designed to manage documents in a digital format. The system helps organizations store, organize, and retrieve important documents electronically instead of using traditional paper files

In many offices, schools, hospitals, and businesses, managing paper documents is difficult, time-consuming, and risky because files can be lost or damaged. This project provides a computerized solution that simplifies document handling and improves efficiency.

The system allows users to upload documents, store them in a structured format, and search for them easily whenever needed. It uses Java programming along with file handling and database concepts to maintain document records securely.

Key features of the system include user authentication, document categorization, fast search functionality, and secure access. Users can perform operations such as adding new documents, viewing existing files, updating details, and deleting unwanted records.

Overall, the Digital Document Record Management System reduces paperwork, saves time and storage space, and ensures safe and efficient document management for organizations.

1.3 Features

User Authentication (Login & Registration)
The system provides secure login and registration so that only authorized users can access and manage documents.

1. **Document Upload Facility**
   Users can upload digital documents (PDF, Word, Images, etc.) and store them safely in the system.
2. **Document Categorization**
   Documents can be organized into categories (such as academic, office, personal) for easy management.
3. **Search and Retrieval**
   Users can quickly search documents using keywords, file name, or category, making retrieval fast and simple.
4. **View and Download Documents**
   Stored documents can be opened, viewed, and downloaded whenever required.
5. **Update Document Details**
   Users can edit document information such as title, description, or category when needed.
6. **Delete Documents**
   Unnecessary or outdated documents can be removed from the system.
7. **Secure Storage**
   The system ensures data safety by storing document details in a structured and protected format.
8. **User-Friendly Interface**
   The application is designed to be simple and easy to use for all types of users.
9. **Time and Space Saving**
   Reduces paperwork and saves physical storage space by maintaining records digitally.

2.PROGRAM ANALYSIS

The Digital Document Record Management System is developed using Java to provide a simple and efficient way to store, manage, and retrieve documents digitally. The program follows a layered structure consisting of the user interface, application logic, and data storage. The user interface accepts input such as login details, document information, and search queries. The application logic processes these inputs and performs operations like validation, storing, updating, searching, and deleting documents. The data storage layer maintains document details using Java file handling or a database such as MySQL or SQLite for permanent storage.

The system uses Object-Oriented Programming concepts like classes and objects to organize the program into modules such as the User Module, Document Management Module, and Search Module. The user first logs into the system, then selects the required operation from the menu. Based on the user's choice, the program processes the request and retrieves or updates the stored data. The system provides outputs in the form of document lists, search results, and confirmation messages. Overall, the program reduces manual paperwork, improves security, and enables quick and easy document access.

2.1 Problem Definition

Problem Definition – Digital Document Record Management System

In many organizations such as schools, colleges, hospitals, and offices, documents are still stored and managed using traditional paper-based methods. This manual system creates several problems including difficulty in storing large volumes of files, risk of document loss or damage, slow searching process, and lack of proper security. Finding a specific document from physical files takes a lot of time and effort, and sharing documents between departments becomes complicated. In addition, maintaining paper records requires large storage space and increases administrative work.

There is a strong need for a computerized solution that can store and manage documents safely in digital form. The system should allow users to upload, organize, search, update, and delete documents easily while ensuring data security and quick access. Therefore, the Digital Document Record Management System is developed using Java to overcome these problems and provide an efficient, secure, and paperless document management solution.

2.2 Project Overview

Project Overview – Digital Document Record Management System (Java)

The Digital Document Record Management System (DDRMS) is a Java-based application developed to manage documents in a digital and organized manner. The main purpose of this project is to replace traditional paper-based record keeping with a computerized system that allows easy storage, retrieval, and management of documents

This project provides features such as user login authentication, document upload, categorization, searching, updating, and deletion of records. The system ensures that only authorized users can access the documents, thereby maintaining security and privacy. It uses Java programming concepts such as object-oriented programming, file handling, and optionally database connectivity to store and manage document details effectively.

The application is designed to be user-friendly and efficient, reducing paperwork and saving time and physical storage space. It can be used in educational institutions, offices, hospitals, and other organizations where document management is essential. Overall, the project demonstrates how Java technology can be used to build a secure and efficient digital document management system.

2.3 Scopes and Limitations

Scope

The Digital Document Record Management System has a wide scope in organizations where document handling is essential. The system can be used in schools, colleges, offices, hospitals, and businesses to store and manage documents digitally. It allows users to upload, search, update, and delete documents easily, which reduces manual paperwork and improves efficiency. The project demonstrates the use of Java programming, file handling, and database connectivity to build a secure and user-friendly document management application. The system can be further expanded by adding cloud storage, mobile access, multi-user support, and advanced security features in the future.

Limitations

Although the system provides basic document management functions, it has some limitations. The project is designed mainly for small-scale use and may not support large organizations with huge volumes of data. Advanced features such as real-time collaboration, cloud backup, and strong encryption are not included in the current version. The system may require manual backup to prevent data loss, and performance may depend on the local system where the application is installed. Future improvements can overcome these limitations by integrating modern technologies.

3.SYSTEM DESIGN

System Design – Digital Document Record Management System (Java)

The Digital Document Record Management System is designed using a three-layer architecture consisting of the presentation layer, application layer, and data storage layer. The presentation layer provides a user-friendly interface where users can log in and perform operations such as uploading, searching, updating, and deleting documents. The application layer contains the business logic that processes user requests, validates inputs, and manages document-related operations. The data layer is responsible for storing document files and their details using Java file handling or a database such as MySQL or SQLite.

The system is divided into modules including the User Authentication Module, Document Management Module, Search Module, and Storage Module. The user authentication module ensures secure access through login verification. The document management module handles adding, viewing, updating, and deleting documents. The search module allows users to find documents quickly using keywords, file names, or categories. The storage module maintains document files and their metadata in an organized manner. The system collects inputs such as user credentials and document details, processes the data, and provides outputs like document lists, search results, and confirmation messages. This design makes the system simple, secure, and easy to maintain and expand.

3.1 Module

1. User Authentication Module

This module manages user registration and login. It verifies the username and password before allowing access to the system. Only authorized users can manage and view documents, ensuring security and privacy.

2. Document Upload Module

This module allows users to upload digital documents such as PDF, Word files, and images. It stores the documents in the system and records their details like title, category, and date.

3. Document Management Module

This is the core module of the system. It enables users to view, update, and delete stored documents. Users can edit document information whenever required.

4. Search Module

This module helps users quickly find documents by using keywords, file names, or categories. It reduces the time required to locate files.

5. Storage Module

This module handles saving and retrieving documents from storage. It uses Java file handling or a database to store document details securely.

6. Report/View Module

This module displays the list of available documents and shows results of operations like upload, update, search, and delete.

4.METHODOLOGY

The Digital Document Record Management System is developed using a structured and step-by-step methodology to ensure the system is simple, secure, and efficient. The development process begins with requirement analysis, where the problems of manual document management and the need for a digital solution are studied. Based on these requirements, the system design is prepared by identifying modules such as user authentication, document management, search, and storage.

After the design phase, the implementation phase is carried out using Java programming language. Object-Oriented Programming concepts such as classes, objects, methods, and encapsulation are used to develop different modules. Java file handling and optional database connectivity are used to store document details and manage files securely.

Once the system is implemented, testing is performed to check whether all functions like login, upload, search, update, and delete are working correctly. Errors are identified and corrected during this stage. Finally, the system is deployed for use, and users can manage documents digitally. This methodology ensures the project is developed in an organized and systematic manner.

4.1. Problem Analysis

Problem Analysis – Digital Document Record Management System

Traditional document management in many organizations still depends on paper-based records. This manual approach creates several challenges in storing, organizing, and retrieving documents efficiently. Large volumes of paper files require significant physical storage space, and managing them becomes difficult as the number of records increases. Documents are also at risk of being misplaced, damaged, or lost due to human error, natural wear, or disasters.Searching for a specific document in a manual system is time-consuming and reduces productivity. Sharing documents between departments or users is also slow and inconvenient. In addition, paper records lack proper security, as unauthorized individuals may access confidential information. Maintaining and updating paper documents increases administrative work and operational costs.

Due to these issues, there is a clear need for a digital solution that can store documents securely, allow quick access, and simplify document management tasks. The Digital Document Record Management System is proposed to address these problems by providing a secure, efficient, and paperless way to manage records.

4.2. System Design

System Design – Digital Document Record Management System (Java)

The Digital Document Record Management System is designed as a structured and modular application to ensure easy usage, security, and scalability. The system follows a three-layer architecture consisting of the presentation layer, application layer, and data storage layer. The presentation layer provides a simple menu-driven interface where users can log in and perform operations such as uploading, searching, updating, and deleting documents. The application layer acts as the core of the system, processing user requests, validating inputs, and controlling all document-related operations. The data storage layer stores document files and their details using Java file handling or a database such as MySQL or SQLite.

The system is divided into major modules including user authentication, document upload, document management, search, and storage. The user authentication module verifies login credentials to ensure only authorized users can access the system. The document upload module allows users to add new documents along with their details. The document management module enables viewing, updating, and deleting documents. The search module helps users quickly find documents using keywords, file names, or categories. The storage module organizes and saves documents securely.

The system accepts inputs such as user credentials and document details, processes them using Java logic, and provides outputs like document lists, search results, and confirmation messages. This design ensures the system is secure, efficient, and easy to maintain and expand in the future.

The program flow and structure are designed so that users can easily interact with the system.

4.3. Program Development

The Digital Document Record Management System was developed using Java by following a step-by-step development process. First, the system requirements were analyzed to understand the problems of manual document management and identify the needed features. Next, the system was designed by dividing it into modules such as user authentication, document upload, document management, search, and storage.

After the design phase, the coding was implemented using Object-Oriented Programming concepts in Java. File handling and optional database connectivity were used to store document details and manage files securely. Once the coding was completed, the system was tested to ensure all functions such as login, upload, search, update, and delete were working correctly. Finally, the application was finalized and made ready for use as a simple and efficient digital document management solution.

4.4. Testing and Validation

The system was tested to ensure all functions such as login, document upload, search, update, and delete work correctly. Errors were identified and fixed during testing. Validation confirmed that the system meets user requirements and produces accurate and secure results.

4.5. Result and Documentation

Result:

The Digital Document Record Management System was successfully developed using Java. The system allows users to securely upload, store, search, update, and delete documents. It reduces paperwork, saves storage space, and improves the speed and efficiency of document management. The application provides a user-friendly interface and ensures secure access through authentication.

Documentation:

The project documentation includes all stages of development such as problem definition, objectives, system design, methodology, program development, testing, and results. It explains how the system works, the technologies used, and the features provided. The documentation serves as a guide for users and future developers to understand, maintain, and enhance the system

4.6. Algorithm

1. Start the program.
2. Display login screen and verify user credentials.
3. Show main menu (Add, View, Search, Update, Delete, Logout).
4. Perform selected document operation.
5. Store or retrieve data from storage.
6. Repeat until user logs out.
7. End the program.

4.7. Flowchart

```
          ( START )
             |
             v
    [ Enter Username & Password ]
             |
             v
        < Is Login Valid? >
         /          \
        No           Yes
        |            |
 [ Show Error Message ]      v
        |       [ Display Main Menu ]
        |            |
        |       < Select Operation >
        |   /   /   /   /   /   \
        | Add  View  Search Update Delete Logout
        |  |   |    |    |    |    |
```

```
|   v    v    v    v    v    v

| [Enter Document Details]
|          |
|       [Save Document]
|          |
|-------------------------------|

    [Display Documents]
          |
|-------------------------------|

    [Enter Keyword]
          |
    [Show Results]
          |
|-------------------------------|

    [Edit Document Details]
          |
    [Save Changes]
          |
|-------------------------------|

    [Delete Document]
          |
|-------------------------------|

     ---- Back to Menu ----
```

|

v

(END)

5.SOURCE CODE

```
import java.util.*;

public class DDRMS {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        ArrayList<String> docs = new ArrayList<>();
        int choice;

        do {
            System.out.println("\n1.Add Document");
            System.out.println("2.View Documents");
            System.out.println("3.Exit");
            System.out.print("Enter choice: ");
            choice = sc.nextInt();
            sc.nextLine();

            if (choice == 1) {
                System.out.print("Enter document name: ");
                String name = sc.nextLine();
                docs.add(name);
                System.out.println("Document Added!");
            }

            else if (choice == 2) {
                System.out.println("Documents List:");
                for (String d : docs)
                    System.out.println(d);
```

```
            }

        } while (choice != 3);

        System.out.println("Program Ended");
    }
}
```

6.INPUT AND OUTPUT SCREENS

INPUT SCREEN

```
1.Add Document
2.View Documents
3.Exit
```

OUTPUT SCREEN

```
Microsoft Windows [Version 10.0.26200.8039]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KGCAS\Desktop\own>javac DDRMS.java

C:\Users\KGCAS\Desktop\own>java DDRMS

1.Add Document
2.View Documents
3.Exit
Enter choice: 1
Enter document name: file
Document Added!

1.Add Document
2.View Documents
3.Exit
Enter choice: 2
Documents List:
file

1.Add Document
2.View Documents
3.Exit
Enter choice: 3
Program Ended

C:\Users\KGCAS\Desktop\own>
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The Digital Document Record Management System was developed to replace manual document handling with a simple digital solution. The system successfully performs basic operations such as adding, viewing, searching, updating, and deleting documents. It improves document organization, reduces paperwork, and provides faster access to records. The project also demonstrates the practical use of Java programming concepts such as file handling, arrays/collections, and menu-driven programming. Although the current version is basic, it provides a strong foundation that can be expanded with advanced features like database integration, cloud storage, and multi-user access.

7.2 Conclusion

The project achieved its main objective of creating a simple and efficient digital document management system. It helps users store and manage documents securely while saving time and storage space. Overall, the system proves that digital solutions can significantly improve document management and can be further enhanced in the future for real-world applications.

8.REFERENCES

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These references were used to understand Java programming concepts and system development.